Volumes of Revolution

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The disk and washer methods**

Find the volume generated when the region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis.

$V = \pi\int_0^4 (\sqrt{x})^2\,dx = \pi\int_0^4 x\,dx = \pi\left[\frac{x^2}{2}\right]_0^4 = 8\pi.$

2 **[Foundational]** • **Target: The disk and washer methods**

The region between $y = x^2$ and $y = 4$ is revolved about the x-axis. Using the washer method, set up and evaluate the volume integral.

Bounds $x = -2$ to 2: $V = \pi\int_{-2}^2 [4^2 - (x^2)^2]dx = \pi\int_{-2}^2 (16 - x^4)\,dx = \pi\left[16x - \frac{x^5}{5}\right]_{-2}^2 = \frac{384\pi}{5}.$

3 **[Foundational]** • **Target: The shell method**

The region under $y = x^2$ on $[0, 2]$ is revolved about the y-axis. Use the shell method to find the volume.

$V = 2\pi\int_0^2 x \cdot x^2\,dx = 2\pi\int_0^2 x^3\,dx = 2\pi\left[\frac{x^4}{4}\right]_0^2 = 8\pi.$

4 **[Foundational]** • **Target: The shell method**

The region bounded by $y = x$, $y = 0$, and $x = 3$ is revolved about the y-axis. Find the volume using the shell method.

$V = 2\pi\int_0^3 x \cdot x\,dx = 2\pi\left[\frac{x^3}{3}\right]_0^3 = 18\pi.$

5 **[Foundational]** • **Target: Comparing methods**

The region under $y = x^2$ on $[0, 2]$ is revolved about the x-axis (not the y-axis). Find the volume using the disk method.

$V = \pi\int_0^2 (x^2)^2\,dx = \pi\int_0^2 x^4\,dx = \pi\left[\frac{x^5}{5}\right]_0^2 = \frac{32\pi}{5}.$

6 **[Foundational]** • **Target: Comparing methods**

The region bounded by $y = \sqrt{x}$ and $y = x$ (between their two intersection points) is revolved about the x-axis. Find the volume using the washer method.

Intersections: $\sqrt{x} = x$ at $x = 0, 1$. On $[0, 1]$, $\sqrt{x} \geq x$, so the outer radius is $\sqrt{x}$, inner is x:
$V = \pi\int_0^1 [(\sqrt{x})^2 - x^2]dx = \pi\int_0^1 (x - x^2)\,dx = \pi\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{\pi}{6}.$

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7 [Foundational] • Target: Volumes about other axes

The region under $y = x^2$ on $[0, 2]$ is revolved about the line $y = -1$. Find the volume, using outer radius $x^2 + 1$.

$V = \pi\int_0^2 [(x^2+1)^2 - 1^2]dx = \pi\int_0^2 (x^4 + 2x^2)\,dx = \pi\left[\frac{x^5}{5} + \frac{2x^3}{3}\right]_0^2 = \pi\left(\frac{32}{5} + \frac{16}{3}\right) = \frac{176\pi}{15}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Volumes about other axes

The region under $y = x^2$ on $[0, 2]$ is revolved about the line $x = 3$ (using the shell method with radius $3 - x$).

$V = 2\pi\int_0^2 (3-x)x^2\,dx = 2\pi\int_0^2 (3x^2 - x^3)\,dx = 2\pi\left[x^3 - \frac{x^4}{4}\right]_0^2 = 2\pi(8-4) = 8\pi$.

9 [Intermediate] • Target: Volumes with trigonometric and exponential curves

The region under $y = \sin x$ on $[0, \pi]$ is revolved about the x-axis. Find the volume, using $\sin^2 x = \frac{1-\cos 2x}{2}$.

$V = \pi\int_0^\pi \sin^2 x\,dx = \pi\int_0^\pi \frac{1-\cos 2x}{2}\,dx = \frac{\pi}{2}\left[x - \frac{\sin 2x}{2}\right]_0^\pi = \frac{\pi}{2}(\pi - 0) = \frac{\pi^2}{2}$.

10 [Intermediate] • Target: Volumes with trigonometric and exponential curves

The region under $y = e^{-x}$ on $[0, 1]$ is revolved about the x-axis. Find the volume, to three decimal places.

$V = \pi\int_0^1 e^{-2x}\,dx = \pi\left[-\frac{1}{2}e^{-2x}\right]_0^1 = \frac{\pi}{2}(1 - e^{-2}) \approx \frac{\pi}{2}(0.8647) \approx 1.358$.

11 [Intermediate] • Target: Setting up without evaluating

Set up (do not evaluate) the volume integral for the region under $y = \ln x$ on $[1, e]$, revolved about the x-axis.

$V = \pi\int_1^e (\ln x)^2\,dx$.

12 [Intermediate] • Target: Setting up without evaluating

Set up (do not evaluate) the volume integral for the region between $y = x^2$ and $y = \sqrt{x}$ (on $[0, 1]$) revolved about the y-axis, using the shell method.

$V = 2\pi\int_0^1 x(\sqrt{x} - x^2)dx$ (outer curve $\sqrt{x}$ minus inner curve x^2, since $\sqrt{x} \geq x^2$ on $[0, 1]$).

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13 **[Intermediate]** • **Target: Visualizing the generating region**

The region under $y = \sqrt{x}$ on $[0, 4]$ (from an earlier disk-method problem) is shown. Sketching this region mentally revolved about the x-axis produces a solid — describe its shape, and confirm the volume 8π found earlier is reasonable by comparing to a cylinder of radius 2 and length 4.

The solid is a paraboloid-like horn, narrow at $x = 0$ and widening to radius 2 at $x = 4$. A full cylinder of radius 2, length 4 would have volume $\pi(2)^2(4) = 16\pi$; since the paraboloid is narrower than the cylinder everywhere except at $x = 4$, a volume of 8π (exactly half the cylinder) is reasonable.

14 **[Intermediate]** • **Target: Volume of a solid with a hole (washer method practice)**

The region between $y = x$ and $y = x^3$ on $[0, 1]$ is revolved about the x-axis. Find the volume using the washer method (outer radius x, inner radius x^3).

$V = \pi\int_0^1 [x^2 - x^6]dx = \pi\left[\frac{x^3}{3} - \frac{x^7}{7}\right]_0^1 = \pi\left(\frac{1}{3} - \frac{1}{7}\right) = \frac{4\pi}{21}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: An applied volume problem** • **Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain**

A vase is modeled by revolving the region under $y = \sqrt{x} + 1$ on $[0, 4]$ about the x-axis, where x is the height (cm) and y is the radius. Find the volume of the vase.

$V = \pi\int_0^4 (\sqrt{x} + 1)^2\,dx = \pi\int_0^4 (x + 2\sqrt{x} + 1)dx = \pi\left[\frac{x^2}{2} + \frac{4}{3}x^{3/2} + x\right]_0^4 = \pi\left(8 + \frac{32}{3} + 4\right) = \pi\left(\frac{68}{3}\right) = \frac{68\pi}{3}\ \text{cm}^3$.

16 **[Challenging]** • **Target: Volume with the shell method, revisited** • **Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals**

The region bounded by $y = 4 - x^2$ and $y = 0$ (for $x \geq 0$) is revolved about the y-axis. Use the shell method to find the volume.

$V = 2\pi\int_0^2 x(4 - x^2)\,dx = 2\pi\int_0^2 (4x - x^3)\,dx = 2\pi\left[2x^2 - \frac{x^4}{4}\right]_0^2 = 2\pi(8 - 4) = 8\pi$.

17 **[Challenging]** • **Target: Comparing methods** • **Cross-Lesson: 4.02 Riemann Sums and Definite Integrals**

Shell method for solid of revolution about y-axis: cylindrical shells have radius $=$.

Final Answer: x. Applying Applications of Integration principles, Shell radius $= x$. Therefore, the verified correct option is x.

18 **[Challenging]** • **Target: Volumes about other axes** • **Cross-Lesson: 5.01 The Fundamental Theorem of Calculus**

Volume of solid of revolution when region $0 \leq y \leq e^{-x^2}$, $0 \leq x \leq 1$ is rotated about y-axis (shell):.

Final Answer: $2\pi\int_0^1 xe^{-x^2}dx$. Shell: $V = 2\pi\int_0^1 xe^{-x^2}dx = \pi(1 - e^{-1})$. Choice C is the setup (correct), choice A = the value.

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19 [Challenging] • Target: Volumes with trigonometric and exponential curves • Cross-Lesson: 6.01 Related Rates

Volume by shell about x-axis: shells have radius $=$.

Final Answer: y. Applying Applications of Integration principles, Shell about x-axis: radius $= y$. Therefore, the verified correct option is y.

20 [Challenging] • Target: Setting up without evaluating • Cross-Lesson: 6.02 Optimization and Curve Sketching

Volume of solid of revolution of $y = \cos x$ about x-axis from 0 to $\pi/2$:.

Final Answer: $\pi^2/4$. $\pi\int_0^{\pi/2} \cos^2 x\,dx = \pi \cdot \pi/4 = \pi^2/4$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Disk Method (Rotation Around Axes)

Find the volume of the solid generated by revolving $y = \sqrt{x}$ from $x = 0$ to $x = 4$ about the x-axis.

a. 8π ✓ **b.** 16π

c. 4π **d.** $32\pi/3$

Disk method: $V = \pi\int_0^4 [f(x)]^2\,dx = \pi\int_0^4 x\,dx = \pi[x^2/2]_0^4 = \pi(16/2) = 8\pi$.

22 [Foundational] • Target: Disk Method (Rotation Around Axes)

Find the volume when the region under $y = x^2$ from $x = 0$ to $x = 2$ is rotated about the x-axis.

a. $\frac{32\pi}{5}$ ✓ **b.** $\frac{16\pi}{5}$

c. 8π **d.** $\frac{64\pi}{5}$

$V = \pi\int_0^2 (x^2)^2\,dx = \pi\int_0^2 x^4\,dx = \pi[x^5/5]_0^2 = \frac{32\pi}{5}$.

23 [Foundational] • Target: Disk Method (Rotation Around Axes)

Find the volume when the region between $x = \sqrt{y}$, $y = 0$, and $y = 4$ is revolved about the y-axis.

a. 8π ✓ **b.** 16π

c. 4π **d.** $32\pi/3$

Disk method with respect to y: $V = \pi\int_0^4 [\sqrt{y}]^2\,dy = \pi\int_0^4 y\,dy = \pi[y^2/2]_0^4 = 8\pi$.

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24 [Foundational] • Target: Washer Method (Hollow Solids)

Find the volume when the region bounded by $y = x$ and $y = x^2$ is revolved about the x-axis.

a. $\frac{2\pi}{15}$ ✓

b. $\frac{\pi}{15}$

c. $\frac{4\pi}{15}$

d. $\frac{\pi}{5}$

Outer $R = x$, inner $r = x^2$. $V = \pi\int_0^1 (x^2 - x^4)\,dx = \pi[x^3/3 - x^5/5]_0^1 = \pi(1/3 - 1/5) = \frac{2\pi}{15}$.

25 [Foundational] • Target: Washer Method (Hollow Solids)

The region bounded by $y = 4 - x^2$ and $y = 0$ is rotated about the line $y = -1$. Set up the outer and inner radii.

a. $R(x) = (4 - x^2) - (-1) = 5 - x^2$, $r(x) = 0 - (-1) = 1$ ✓

b. $R(x) = 4 - x^2$, $r(x) = 1$

c. $R(x) = 5 - x^2$, $r(x) = 0$

d. $R(x) = 4 - x^2 + 1$, $r(x) = 2$

Distance from curve to axis of revolution $y = -1$: outer radius $R(x) = (4 - x^2) - (-1) = 5 - x^2$, inner radius $r(x) = 0 - (-1) = 1$.

26 [Foundational] • Target: Washer Method (Hollow Solids)

Evaluate the volume in the previous problem rotated about $y = -1$ on $[-2, 2]$.

a. $\frac{384\pi}{15} = \frac{128\pi}{5}$ ✓

b. $\frac{64\pi}{5}$

c. 32π

d. $\frac{256\pi}{5}$

$V = \pi\int_{-2}^{2}[(5 - x^2)^2 - 1^2]\,dx = 2\pi\int_0^2 (24 - 10x^2 + x^4)\,dx = 2\pi[24x - \frac{10}{3}x^3 + \frac{x^5}{5}]_0^2 = 2\pi(48 - 80/3 + 32/5) = 2\pi(\frac{720 - 400 + 96}{15}) = 2\pi(\frac{416}{15})$ wait: $24(2) = 48$; $2\pi(48 - 80/3 + 32/5) = 2\pi(\frac{416}{15}) = \frac{832\pi}{15}$.

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27 [Foundational] • Target: Cylindrical Shells Method

State the cylindrical shells formula for a region under $y = f(x)$ on $[a, b]$ revolved about the y-axis.

a. $V = 2\pi\int_a^b xf(x)\,dx$ ✓

b. $V = \pi\int_a^b [f(x)]^2\,dx$

c. $V = 2\pi\int_a^b [f(x)]^2\,dx$

d. $V = \pi\int_a^b x^2 f(x)\,dx$

Circumference $= 2\pi x$, height $= f(x)$, thickness $= dx$. Shell volume $= 2\pi\int_a^b xf(x)\,dx$.

28 [Foundational] • Target: Cylindrical Shells Method

Find the volume when the region under $y = 3x - x^2$ from $x = 0$ to $x = 3$ is revolved about the y-axis.

a. $\frac{27\pi}{2}$ ✓

b. 27π

c. $\frac{81\pi}{4}$

d. $\frac{9\pi}{2}$

Shells: $V = 2\pi\int_0^3 x(3x - x^2)\,dx = 2\pi\int_0^3 (3x^2 - x^3)\,dx = 2\pi[x^3 - \frac{x^4}{4}]_0^3 = 2\pi(27 - 81/4) = 2\pi(27/4) = \frac{27\pi}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Cylindrical Shells Method

Use cylindrical shells to find the volume when $y = e^{-x^2}$ on $[0, 1]$ is revolved about the y-axis.

a. $\pi(1 - e^{-1})$ ✓

b. $2\pi(1 - e^{-1})$

c. πe^{-1}

d. $\frac{\pi}{2}(1 - e^{-1})$

$V = 2\pi\int_0^1 xe^{-x^2}\,dx$. Let $u = -x^2$, $du = -2x\,dx$. $V = -\pi\int_0^{-1} e^u\,du = \pi\int_{-1}^0 e^u\,du = \pi[e^u]_{-1}^0 = \pi(1 - e^{-1})$.

30 [Intermediate] • Target: Cross-Sections with Known Geometric Shapes

The base of a solid is the circle $x^2 + y^2 = 4$. Cross-sections perpendicular to the x-axis are squares. Find the volume.

a. $\frac{128}{3}$ ✓

b. 16π

c. 32

d. $\frac{64}{3}$

Side of square $s = 2y = 2\sqrt{4 - x^2}$. Area $A(x) = s^2 = 4(4 - x^2)$.
$V = \int_{-2}^2 4(4 - x^2)\,dx = 8\int_0^2 (4 - x^2)\,dx = 8[4x - x^3/3]_0^2 = 8(8 - 8/3) = 8(16/3) = 128/3$.

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31 [Intermediate] • Target: Cross-Sections with Known Geometric Shapes

The base of a solid is bounded by $y = x^2$ and $y = 4$. Cross-sections perpendicular to the y-axis are semicircles. Find the volume.

a. 4π ✓

b. 2π

c. 8π

d. $\frac{8\pi}{3}$

Diameter $d = 2x = 2\sqrt{y} \Rightarrow r = \sqrt{y}$. Area $A(y) = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi y$. $V = \int_0^4 \frac{1}{2}\pi y\, dy = \frac{\pi}{2}[y^2/2]_0^4 = \frac{\pi}{2}(8) = 4\pi$.

32 [Intermediate] • Target: Cross-Sections with Known Geometric Shapes

The base is the triangle with vertices $(0, 0), (1, 0), (0, 1)$. Cross-sections perpendicular to the x-axis are equilateral triangles. Find the volume.

a. $\frac{\sqrt{3}}{12}$ ✓

b. $\frac{\sqrt{3}}{6}$

c. $\frac{1}{6}$

d. $\frac{\sqrt{3}}{4}$

Side $s = 1 - x$. Area of equilateral triangle $A(x) = \frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}(1 - x)^2$.
$V = \frac{\sqrt{3}}{4}\int_0^1 (1 - x)^2\, dx = \frac{\sqrt{3}}{4}[-\frac{(1-x)^3}{3}]_0^1 = \frac{\sqrt{3}}{4}(1/3) = \frac{\sqrt{3}}{12}$.

33 [Intermediate] • Target: Cylindrical Shells Method

Volumes: for the same region rotated, which is larger: shell or washer?

a. Shell is always larger

b. **They give the same volume** ✓

c. Washer is always larger

d. Depends on axis and region

Both methods give the same volume for the same solid.

34 [Intermediate] • Target: Cross-Sections with Known Geometric Shapes

Volume of solid with square cross-sections perpendicular to x-axis, bounded by $y = \sqrt{x}$ and $x = 4$:

a. 4

b. 8 ✓

c. 16/3

d. 8/3

$A(x) = (\sqrt{x})^2 = x$; $V = \int_0^4 x\, dx = 8$.

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35 [Intermediate] • Target: The disk and washer methods

Volume of solid with circular cross-sections of radius $f(x)$:

a. $\int_a^b \pi f(x)dx$

b. $\int_a^b f(x)^2 dx$

c. $\int_a^b \pi[f(x)]^2 dx$ ✓

d. $2\pi\int_a^b f(x)dx$

Cross-section area $= \pi r^2 = \pi[f(x)]^2$.

36 [Intermediate] • Target: The shell method

Which gives volume by the disk method about the y-axis?

a. $\int_a^b f(y)\,dy$

b. $\pi\int_a^b [f(x)]^2\,dx$

c. $2\pi\int_a^b xf(x)\,dx$

d. $\pi\int_c^d [f(y)]^2\,dy$ **(solve for x first)** ✓

Rotate about y-axis; use $x = f(y)$.

37 [Intermediate] • Target: Comparing methods

Volume of the solid of revolution of $y = \sqrt{9 - x^2}$ (semicircle) about x-axis:

a. 18π

b. $\frac{4}{3}\pi \cdot 27 = 36\pi$

c. 36π ✓

d. $\frac{4}{3}\pi(3)^3 = 36\pi$

$V = \pi\int_{-3}^{3}(9 - x^2)dx = 36\pi$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Volumes about other axes • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Volume of revolution of $y = x^3$ about x-axis from 0 to 1:

a. π

b. $\pi/5$

c. $\pi/7$ ✓

d. $3\pi/7$

Applying Applications of Integration principles, $V = \pi\int_0^1 x^6\,dx = \pi/7$. Therefore, the verified correct option is $\pi/7$.

39 [Challenging] • Target: Volumes with trigonometric and exponential curves • Cross-Lesson: 6.01 Related Rates

The region under $y = \sin(x)$ for $0 \le x \le \pi$ is revolved about the x-axis. Find the volume.

a. $\frac{\pi}{2}$

b. π^2

c. $\frac{\pi^2}{2}$ ✓

d. $2\pi^2$

Disk method: $V = \pi\int_0^{\pi} \sin^2(x)\,dx = \frac{\pi^2}{2}$.

40 [Challenging] • Target: Setting up without evaluating • Cross-Lesson: 6.02 Optimization and Curve Sketching

The region under $y = e^x$ for $0 \le x \le \ln(2)$ is revolved about the x-axis. Find the volume.

a. $\frac{3}{2}$

b. 3π

c. $\frac{3\pi}{2}$ ✓

d. $2\pi(-1 + \ln(4))$

Disk method: $V = \pi\int_0^{\ln(2)} e^{2x}\,dx = \frac{3\pi}{2}$.

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41 [Challenging] • Target: Visualizing the generating region • Cross-Lesson: 7.01 Integration by Substitution

The region under $y = e^x$ for $0 \le x \le \ln(5)$ is revolved about the x-axis. Find the volume.

a. 12

b. 12π ✓

c. 24π

d. $2\pi(-4 + \ln(3125))$

Disk method: $V = \pi\int_0^{\ln(5)} e^{2x}\,dx = 12\pi$.

42 [Challenging] • Target: Volume of a solid with a hole (washer method practice) • Cross-Lesson: 7.02 Integration by Parts

The region under $y = \sqrt{x}$ for $0 \le x \le 3$ is revolved about the x-axis. Find the volume.

a. $\frac{9}{2}$

b. $\frac{9\pi}{2}$ ✓

c. 9π

d. $\frac{36\sqrt{3}\pi}{5}$

Disk method: $V = \pi\int_0^3 x\,dx = \frac{9\pi}{2}$.

43 [Challenging] • Target: An applied volume problem • Cross-Lesson: 8.01 Area Between Curves

The solid whose cross-sections perpendicular to the x-axis are equilateral triangles with side $f(x)$ has:

a. $V = \frac{\sqrt{3}}{4}\int f^2\,dx$ ✓

b. $V = \int f\,dx$

c. $V = \pi\int f^2\,dx$

d. $V = 3\int f\,dx$

Area of equilateral triangle side s: $\frac{\sqrt{3}}{4}s^2$.

44 [Challenging] • Target: Volume with the shell method, revisited • Cross-Lesson: 8.03 Arc Length and Improper Integrals

The region under $y = \sqrt{x}$ for $0 \le x \le 4$ is revolved about the x-axis. Find the volume.

a. 8

b. 8π ✓

c. 16π

d. $\frac{128\pi}{5}$

Disk method: $V = \pi\int_0^4 x\,dx = 8\pi$.

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45 [Challenging] • Target: Disk Method (Rotation Around Axes) • Cross-Lesson: 9.01 Separable Differential Equations

Volume by shells revolving $y = x^2$ about the y-axis from 0 to 1:

a. $\pi/2$ ✓

b. $\pi/4$

c. π

d. $2\pi/4$

$V = 2\pi\int_0^1 x \cdot x^2\,dx = 2\pi[x^4/4]_0^1 = \pi/2.$

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